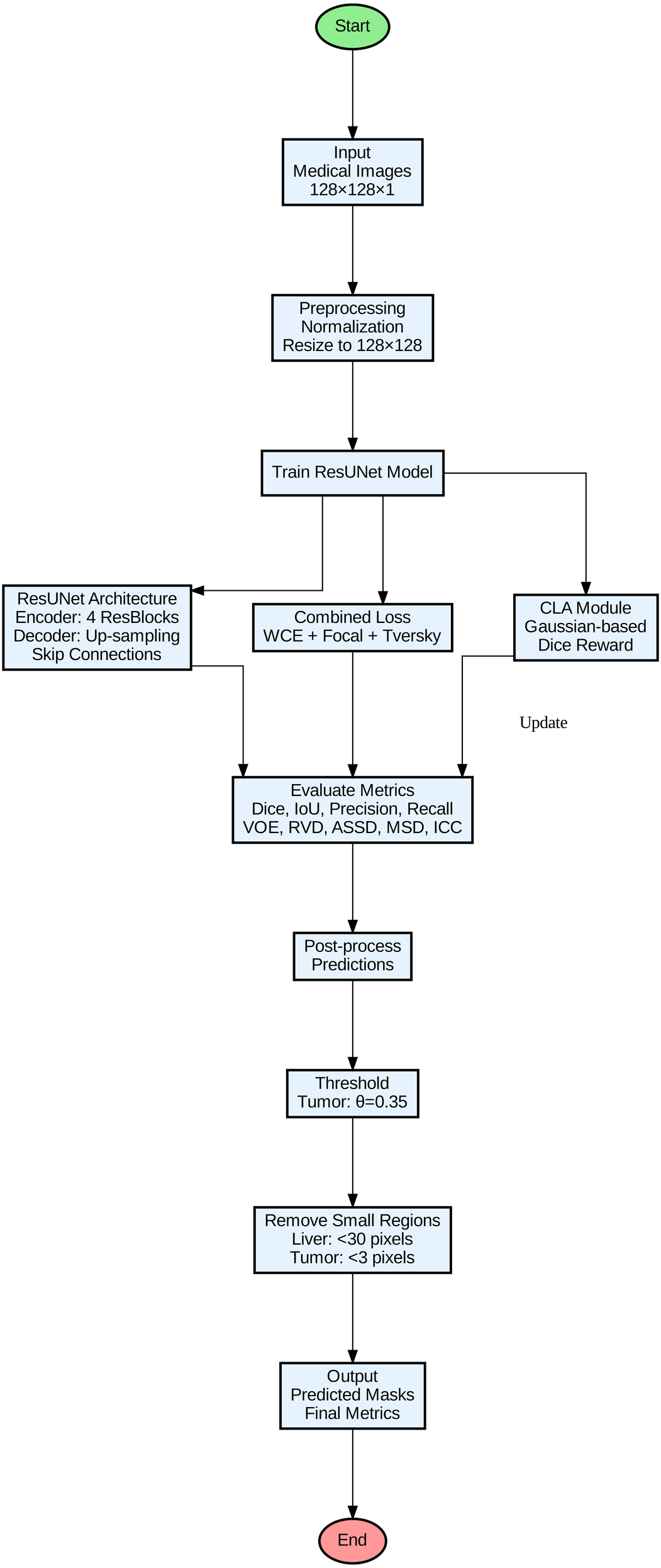
Start
Input
Medical Images
128×128×1
Preprocessing
Normalization
Resize to 128×128
Train ResUNet Model
ResUNet Architecture
Encoder: 4 ResBlocks
Decoder: Up-sampling
Skip Connections
Combined Loss
WCE + Focal + Tversky
CLA Module
Gaussian-based
Dice Reward
Update
Evaluate Metrics
Dice, IoU, Precision, Recall
VOE, RVD, ASSD, MSD, ICC
Post-process
Predictions
Threshold
Tumor: θ=0.35
Remove Small Regions
Liver: <30 pixels
Tumor: <3 pixels
Output
Predicted Masks
Final Metrics
End